

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [2]: iris = load_iris()
```

```
In [3]: print(iris.keys())
```

```
dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names', 'filename', 'data_module'])
```

```
In [4]: df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target
df.head()
```

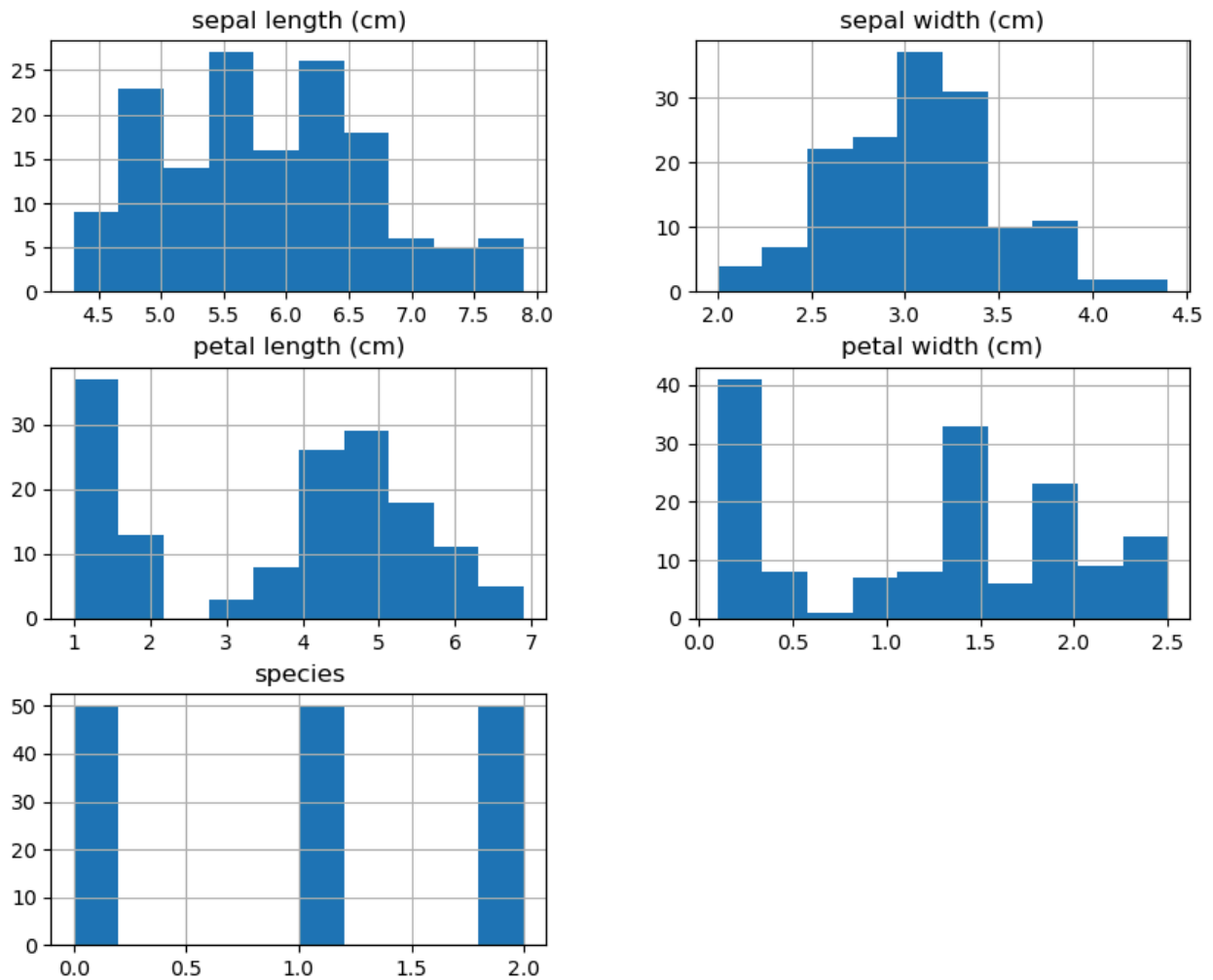
```
Out[4]:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
In [5]: df.shape
```

```
Out[5]: (150, 5)
```

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In [6]: df.hist(figsize=(10,8))
plt.show()
```



```
In [7]: X = df.drop('species', axis=1)
        y = df['species']
```

```
In [8]: X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
```

```
In [9]: model = LogisticRegression(max_iter=200)
        model.fit(X_train, y_train)
```

```
Out[9]: LogisticRegression ⓘ ?
        Parameters
```

```
In [10]: y_pred = model.predict(X_test)
```

```
In [11]: accuracy = accuracy_score(y_test, y_pred)
        print("Accuracy:", accuracy)
```

Accuracy: 1.0

```
In [12]: cm = confusion_matrix(y_test, y_pred)
        print(cm)
```

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

```
In [13]: sample = [[5.1, 3.5, 1.4, 0.2]]  
         prediction = model.predict(sample)  
         print("Predicted class:", iris.target_names[prediction])
```

Predicted class: ['setosa']

C:\Users\ashish kumar\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
 warnings.warn(

In []: